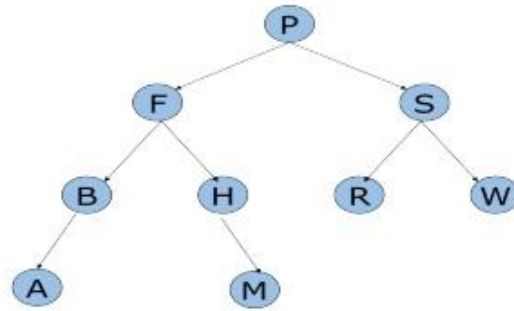
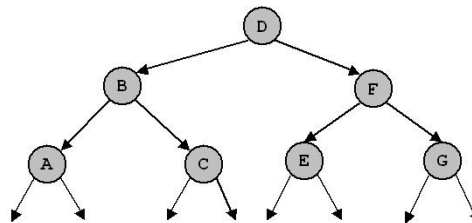


Lab 10: Trees



1. Given the tree above:
 - a. What is the length of the path from P → H?
 - b. What is the height of the tree given above?
 - c. What is the degree of node S?
 - d. What is the degree of the tree given above?
 - e. Write the preorder, inorder, postorder, and level order traversals of the binary tree given below:
2. Write the preorder, inorder, postorder, and level order traversals of the tree given below:



- What is the length of the path from F → E?
 - What is the height of the tree given?
 - What is the degree of node C?
 - What is the degree of the tree given?
3.

```

graph TD
    45((45)) --> 25((25))
    45 --> 55((55))
    25 --> 16((16))
    25 --> 35((35))
    35 --> 32((32))
    55 --> 61((61))
  
```

 1. What is the length of the path from 45 → 32?
 2. What is the height of the tree given?
 3. What is the degree of node 35?
 4. What is the degree of the tree given?
 5. Write the preorder, inorder, postorder, and level order traversals of the binary tree.
 4. Write a C++ program that builds a binary tree and displays the postorder, inorder, and preorder traversals of the tree. Your program should be implemented using an array.